

Lesson 9.2: Equilibrium: zero-sum games

Game theory

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Best response and equilibrium: zero-sum games

	b_1	b_2	b_3	b_4	min
a_1	0	1	7	6	0
a_2	4	2	3	4	2
a_3	3	1	0	2	0
a_4	0	0	7	3	0
Max	4	2	7	6	

- ▶ each strategy in play (a_2 , b_2) is best response to the other
- ▶ paired best responses called an *equilibrium*
- ▶ in zero-sum games, equilibria are matching *Maximin* strategies

Existence

- ▶ not all games have equilibria

	b_1	b_2
a_1	2*	0*
a_2	1*	3*

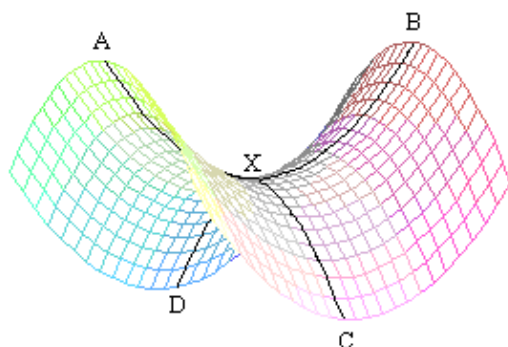
	b_1	b_2	min
a_1	2	0	0
a_2	1	3	1
Max	2	3	

	b_1	b_2	b_3
a_1	1*	2	6*
a_2	7*	3*	3*
a_3	3	2*	5

	b_1	b_2	b_3	min
a_1	1	2	6	1
a_2	7	3	3	3
a_3	3	2	5	2
Max	7	3	6	

Saddle points

- ▶ in zero-sum games, equilibria are called *saddle points*



	b_1	b_2	b_3	min
a_1	1	2	6	1
a_2	7	3	3	3
a_3	3	2	5	2
Max	7	3	6	

- ▶ saddle points have outcomes with payoffs:
 - ▶ minimal in their row; and
 - ▶ maximal in their column